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Supply chain risks in agri-food systems: a comprehensive review of economic vulnerabilities and mitigation approaches

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The global agri-food system (AFS) is increasingly vulnerable to a complex web of economic, environmental, and geopolitical disruptions. This review paper critically examines the economic vulnerabilities embedded within agri-food supply chain (AFSC), focusing particularly on smallholder farmers, export-oriented economies, and the broader risks associated with globalization. Drawing on recent crises such as the COVID-19 pandemic and the Russia–Ukraine conflict, the paper explores how systemic shocks disrupt production, distribution, and consumption, leading to increased food insecurity, especially in the Global South. Key issues include limited financial access, infrastructural deficits, digital exclusion, and food price volatility. The paper highlights a range of mitigation strategies, including policy reform, digital technology adoption (e.g., blockchain, internet of things), local food system strengthening, financial risk transfer instruments, and collaborative capacity building. Through global case studies and critical analysis, the paper identifies persistent research gaps—particularly regarding informal food systems and the contextual adaptability of technological innovations. It calls for interdisciplinary approaches and multi-stakeholder cooperation to foster resilient, inclusive, and sustainable AFSs capable of withstanding future shocks. Moreover, this paper advances key Sustainable Development Goals by protecting smallholder livelihoods (SDG 1 and 2), promoting digital agriculture and infrastructure (SDG 9), improving supply chain transparency (SDG 12), and addressing climate risks with adaptive strategies (SDG 13). It lays a foundation for resilient and sustainable AFSs through policy and innovation.

KEYWORDS

agri-food supply chain, food system resilience, SDGs, risk mitigation, food security

1 Introduction

The agri-food system (AFS) is composed of diverse and intricate activities ranging from sustainable food production to its responsible disposal. Hence, it involves all the intermediate stages through which food is generated, refined, delivered and ultimately utilized (Vallejo-Rojas et al., 2015; Borsellino et al., 2020). However, AFSs are constantly evolving and becoming more and more complex in response to several factors such as globalization and technological innovations (Charatsari et al., 2022; Cricelli et al., 2024). The evolution of AFSs from local and subsistence-based to global and commercial-based is truly remarkable. This incredible transformation is heavily dependent on increased productivity and efficiency due to digitalization (Requier-Desjardins et al., 2003; Carmela Annosi et al., 2020). Now, it is possible

to consume food produced from other distantly located geographic regions involving complex agri-food supply chains (AFSCs) and passing through diverse regulatory frameworks. This global reach has fortified food security and produced new opportunities for producers (Irani and Sharif, 2016; Alabi and Ngwenyama, 2023). However, on the other hand, intricate structure of food networks rendered it prone to diverse vulnerabilities. Climate change, geopolitical tensions, labor shortages, trade restrictions, infrastructure weaknesses, market volatility, and technological disruptions can all introduce vulnerabilities at various points in AFSC. The effects of even a small disruption can be enormous (Ali et al., 2023; Mkumbukiyi et al., 2025). Thus, it is crucial to study these economic risks to support global food security and the Sustainable Development Goals of Zero Hunger (SDG 2); Responsible Consumption and Production (SDG 12) and Climate Action (SDG 13).

Considerable amount of global labor force is associated with agriculture. This percentage is notably elevated in developing nations where agriculture serves as the principal source of subsistence, especially in rural regions (Blanco and Raurich, 2022; Mohsin et al., 2024). Furthermore, agriculture utilizes land and worsens environmental problems like soil erosion, deforestation, and greenhouse gases, which make sustainability harder to achieve (Francaviglia et al., 2023; Singh et al., 2024). AFSCs play a crucial role in achieving the Sustainable Development Goals (SDGs) by directly influencing food security, sustainable agriculture, and climate-related action. For instance, SDG 2, depend on efficient and resilient AFSCs to ensure food availability and access. Similarly, SDGs 12 and 13, which focus on responsible consumption and production and climate action, are deeply impacted by the way AFSCs manage waste, resources, and reduce emission in food production and distribution (El Bilali et al., 2021; Sannou et al., 2023). AFSCs involve a lot of different players, from small-scale farmers to huge multinational companies, and each faces their own set of challenges and limitations. For instance, smallholder farmers often have trouble getting access to credit, technology, or even proper market info. On the other hand, large corporations struggle with the complexity of their supply chains, staying compliant with regulations, and navigating market shifts. These challenges are distinct, but for AFSCs to function effectively, collaboration is key. For this, each group may need tailored solutions, but cooperation is essential (Williams et al., 2023). Recent years have seen intense disturbances in the distribution network resulting from the Covid-19 pandemic, threatening food security and causing economic disruption globally (Nicola et al., 2020; Alabi and Ngwenyama, 2023). The Middle East and North Africa regions have seen grain price spikes due to geopolitical instability between Russia and Ukraine (Kozielec et al., 2024). Resilient supply chains can withstand acute or chronic disruptions and are more adaptable to long-term systemic changes. Thus, ensuring robustness of distribution networks has become a primary concern of the governments (Katsaliaki et al., 2022; Nakandala et al., 2025).

AFSCs play a key role in ensuring food security, the health of soils is still the foundation for long-term food production. A balanced approach, that acknowledges both sustainable land management (SDG 15) and efficient supply chains (SDG 2), is essential for achieving true global food security. Producers, manufacturers, sellers, buyers, and consumers are intricately interwoven throughout these supply networks (Cook et al., 2011; Cirone et al., 2023). A single failure in the trade corridor can really cause disruptions with AFSC (Katsaliaki

et al., 2022; Gorton et al., 2006). Small household farmers, in particular, deal with all kinds of challenges, like limited access to credit, market info, and reliable irrigation systems. On top of that, delays or disruptions in getting things like fertilizers, seeds, and pesticides can seriously disrupt food production. When these key inputs are missing or late, it's not just the farmers who suffer rather the whole AFS is affected (Rao et al., 2017; Balana et al., 2022). Early warning systems coupled with climate smart agriculture can enhance the robustness of distribution networks (van Ginkel and Biradar, 2021). Studies also suggest the adoption of tools like satellite imaging, traceability, blockchain, and digital monitoring to cope with supply chains risks (Sengupta et al., 2022; Bosona and Gebresenbet, 2023; Wang et al., 2024).

The resilient AFSC is crucially important in this modern era. However, published literature does not present a comprehensive picture of the economic vulnerabilities as well as their mitigation. Most of the studies discuss individual disruptions ignoring their interconnectedness (Ackleson and Kastner, 2011; Ali et al., 2023; Dovbischuk, 2023). Since, AFSCs have evolved at a global scale so through understanding of their economic vulnerabilities and their mitigation aspects are of prime concern to the world in order to achieve SDGs. These studies hold significant importance in developing countries, where AFSCs are more vulnerable to disruptions and there is a scarcity of scientific research to suggest mitigation strategies (Kumar and Agrawal, 2023; Hasan et al., 2023). Furthermore, the use of technological innovations in order to achieve resilient AFSCs is also insufficient (Joshi et al., 2023). The study was undertaken by collecting and analyzing published academic papers, reports and other reliable information using a review approach. The literature reviewed captures risks in AFSCs their economic impacts and possible ways to mitigate them. This study highlights supply chain resilience concerns and solutions using agricultural insights, economic theory, and management ideas.

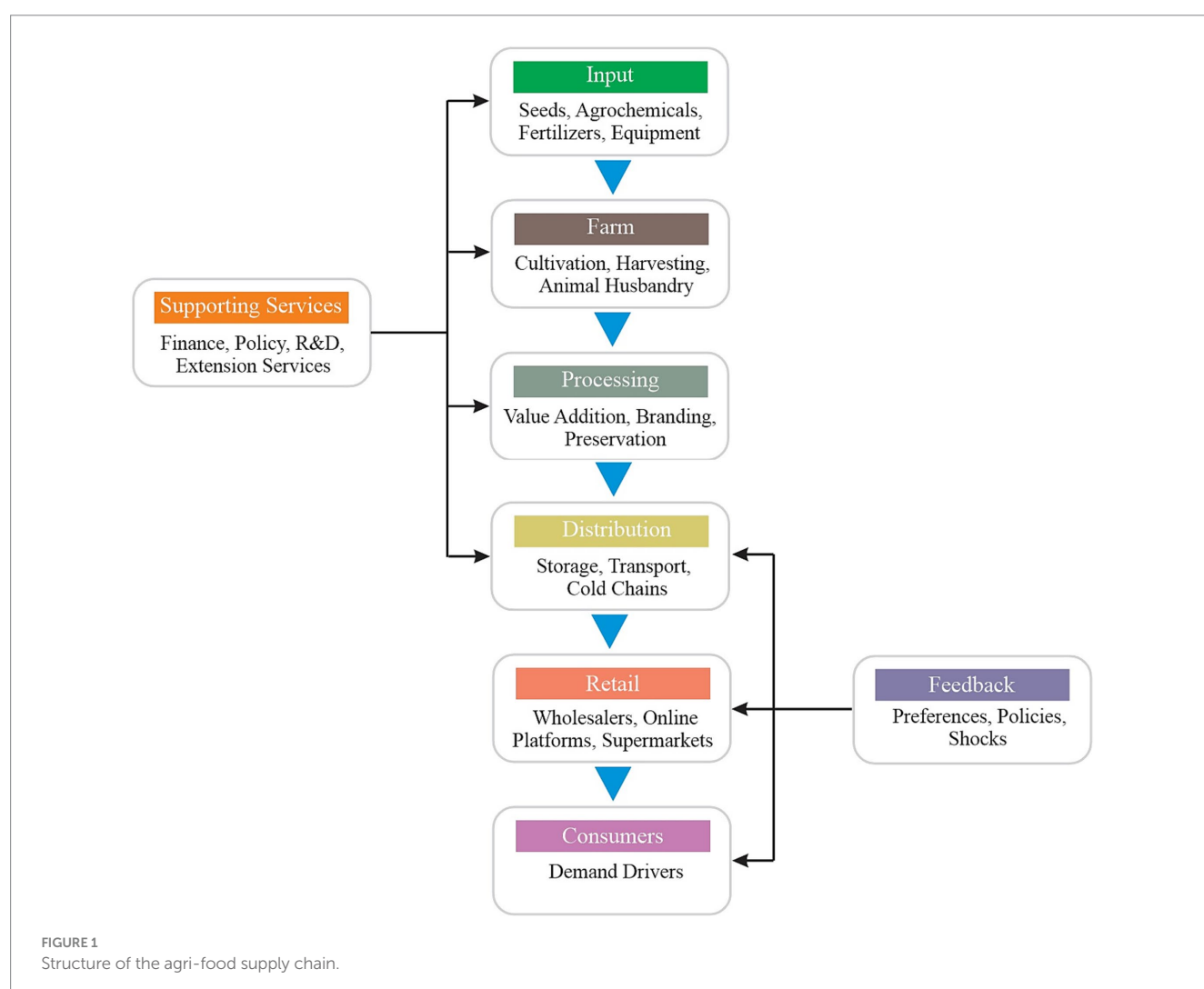
This study adopts an integrative approach by bridging insights gained from AFSC operations, agriculture economics, and risk management. This review aims to address the following questions: What are the main economic risks impacting AFSC? What are the impacts of these risks on various countries and groups? What kind of disruptions, like weather events, political issues, or labor shortages, cause the most problems to supply chain? What techniques, including new tools like blockchain and satellite surveillance, are able to reduce these risks and strengthen supply chains? The remainder of this study is split into eight major sections offering a thorough investigation of supply chain risks in AFSs. Section 2 outlines the structure and complexity of AFSCs, discussing their stages, inherent characteristics, and the implications of globalization and lean system designs. Section 3 classifies the various types of risks affecting these systems, including climate-related, economic, logistical, labor, regulatory, technological, and social risks. Section 4 delves into the economic vulnerabilities of AFSs, particularly focusing on smallholder farmers, export-dependent economies, and infrastructure weaknesses in developing regions. Section 5 presents a range of mitigation and risk management strategies, such as policy interventions, financial tools, technological innovations, and supply chain redesign. Section 6 provides case studies of recent disruptions, such as the COVID-19 outbreak, the Ukraine-Russia conflict, and environmental challenges, illustrating real-world consequences of supply chain fragility. Section 7 identifies major deficiencies in the literature and suggests pathways for

upcoming interdisciplinary studies. Finally, Section 8 wraps up with an overview of findings and provides policy suggestions to enhance the resilience and sustainability of global AFSCs.

2 Structure and complexity of AFSCs

The first stage in AFSC is the procurement of input materials, (Stage 1—Inputs; Figure 1). This includes seeds, fertilizers, agrochemicals, and farming equipment, which serve as the foundation for agricultural production. The second stage (Stage 2—Farm) is the food production stage, involving cultivation, harvesting, and animal husbandry. This stage corresponds to “Farm” in Figure 1 and plays a central role in determining the overall sustainability of the supply chain. Environmental and social aspects such as soil health, water consumption, fertilizer emissions, biodiversity conservation, and fair labor practices, are key considerations at this stage. Technology also contributes to improving agricultural efficiency and minimizing resource waste through precision farming and smart monitoring systems. The third stage (Stage 3—Processing) is the production of finished products through processing, value addition, and packaging of the raw materials. One crucial aspect of this stage is to align balance between demand and supply (Rathor et al., 2022). Technology is

crucial for increasing production efficiencies and decreasing resource waste (Chiang et al., 2021). The fourth stage (Stage 4—Distribution) is the transportation of finished products from production sites to the warehouses for temporary storage or timely distribution (Rathor et al., 2022). As indicated in Figure 1, this stage encompasses storage, transport, and cold-chain logistics. This stage considers environmental, social, and economic factors to achieve sustainable as well as responsible supply chain characteristics (Hou et al., 2015). For example, environmental dimensions may include reducing carbon emissions from transportation, improving energy efficiency, and adopting cleaner logistics solutions. Inventory management is crucial for reducing warehouse holding costs and efficient order delivery. The fifth stage (Retail and Consumers) involves order fulfillment. This stage involves picking and shipping products to either retailers or consumers. The speed and accuracy of this stage greatly impacts consumer satisfaction (Rathor et al., 2022). It also connects closely to consumer feedback mechanisms and demand drivers, as reflected in Figure 1. Advanced tools like internet of things (IoT), artificial intelligence (AI), and blockchain help minimize disruptions in the distribution network. These technologies also support earlier stages, such as farming and processing, by improving traceability, monitoring production efficiency, and ensuring sustainability compliance. IoT enables real-time tracking to prevent losses, AI optimizes



decision-making and inventory management, though it struggles with sudden demand spikes. The blockchain ensures transparency and security, despite potential vulnerabilities. Together, these technologies can enhance supply chain efficiency as well as its robustness (Katsaliaki et al., 2022; Moberg et al., 2004).

Since, food products are perishable therefore they need to be efficiently handled and properly stored to sustain their quality. Moreover, seasonal demand fluctuations make pricing strategies and inventory management a complicated task. Natural disasters arise food safety issues and disrupt supply chains (Kuizina et al., 2023). Researchers have proposed optimization models to deal with such risks (Taşkın and Bilgen, 2021). Furthermore, the adoption of advanced tools like AI, IoT, and blockchain is helping to enhance the robustness of the supply network (Dadi et al., 2021; Menon and Jain, 2024). Supply chains have shifted from a local to global scale interconnecting multiple actors through complex relationships (Menon and Jain, 2024). This transformation has also expanded sustainability considerations to include carbon credits, energy efficiency, and reductions in transportation-related emissions across all stages of the supply chain. This transformation of the supply chains confers both challenges as well as opportunities. They face a range of risks including traceability, imbalance between supply and demand, and food security and its safety. However, digital technologies are facilitating to cope with these risks efficiently (Dadi et al., 2021). Integration of local actors, such as small-scale farmers, into the complex and evolving global supply chains is challenging particularly for countries having large populations like China (Zhang and Aramyan, 2009).

The upsurge of global value chains promises efficient trade in food products. These chains integrate AFSC by the fragmentation of products on the international scale (Katsaliaki et al., 2022; Xiao et al., 2017). The worldwide value chains have made it possible for emerging nations to participate in international markets and boost their economy (Del Prete et al., 2017; Mao, 2021). Global AFSCs contribute to environmental problems, like deforestation, soil degradation, and greenhouse gas emissions due to resource-heavy production and long-distance transportation. They also create economic disparities, with small farmers often getting lower compensation, while larger corporations benefit from economies of scale, leading to market imbalances and food insecurity (Sun, 2018). Application of lean principles to streamline supply chain is gaining popularity. These principles also help to reduce waste. However, by applying these principles, supply chain become more prone to vulnerabilities raising concerns about their resilience (Taylor, 2006; Krstic et al., 2023). Thus, achieving balance between the distribution network efficiency and its robustness presents a challenge to the managers (Joshi et al., 2023). Implementation of block chain technology can help to achieve efficiency and traceability in agri-food distribution networks (Chen et al., 2021; Toader et al., 2024).

3 Classification of supply chain risks

3.1 Environmental and climate-related risks

Environmental factors such as droughts, heatwaves, and floods significantly affect agricultural productivity and the agricultural food supply chain (AFSC) (Malik et al., 2022; Ali et al., 2023). Droughts

reduce soil moisture, leading to crop failures, higher market prices, and lower farm incomes. These economic challenges are further exacerbated by intermediaries capturing a substantial share of value and farmers' dependence on costly agricultural inputs. Although drought-resistant crop varieties can mitigate some adverse effects, their performance still depends on maintaining healthy soils. Extreme weather events and soil degradation further undermine resilience, underscoring the importance of sustainable soil management (Kundzewicz et al., 2002; Renard et al., 2023). According to Khan et al. (2020), rice and wheat production in Pakistan are particularly vulnerable to climate-related risks, which could result in a projected GDP loss of USD 19.5 billion by 2050. Extreme weather events also damage transportation infrastructure, causing logistical delays and supply disruptions. Climate-induced changes are expected to reduce crop yields by approximately 23% by 2050 and increase price volatility (Haile et al., 2017). This decline in production could affect global GDP by up to 4.6% by 2060 (Tan et al., 2024). Rising agricultural commodity prices can also reduce domestic consumption, particularly in urban areas (Khan et al., 2020). This not only strains household budgets but also disrupts key dimensions of food security, particularly availability and access. When prices surge, food accessibility becomes limited for many households, and production slowdowns further constrain overall food availability. Consequently, the efficiency of supply chains deteriorates, intensifying food insecurity in regions already facing limited access to essential resources (Malik et al., 2022).

3.2 Economic and market risks

Price volatility of the agri-food products negatively affects their supply, particularly from developing countries. In countries with poor infrastructure and high inflation, price instability leads to reduced production (Subervie, 2008). Price volatility increases production costs, especially for fertilizers, pesticides, and seeds, making it harder for farmers to sustain production. These rising input costs limit their ability to invest in essential resources, leading to reduced productivity and inefficiencies through AFSCs. In order to mitigate price volatility risk, firms can employ scientific approaches such as real options valuation and total cost of ownership (Gaudenzi et al., 2020). Disruptions like COVID-19 can cause input shortages, affecting production and making agricultural products more costly (Hobbs, 2021). Systematic issues in agri-food logistics, coupled with poor resource management, have resulted in over 33% of food losses, exacerbating global food insecurity (Joshi et al., 2023). Food waste reduces food availability, drives up prices, and limits access to nutrients for vulnerable groups. Moreover, complex supply chains complicate food traceability and management, further exacerbating these challenges (Chen et al., 2021). This absence of transparency can result in food safety problems, fraudulent behavior as well as loss of consumer trust (Menon and Jain, 2024). The loss of consumer confidence in food supply can disrupt market demand and market access, while the imbalance and vertical integration of market forces may lead to unfair conditions for suppliers and limit the participation of smallholders within the distribution network (Young and Hobbs, 2002). The COVID-19 crisis and additional disruptions revealed weaknesses in the agricultural food supply chains, reduced productivity and diversity, and worsened access problems in poorly integrated and less developed areas (Swinen and Vos, 2021).

Solutions for this include the use of circular economy principles (Borrello et al., 2016) and blockchain technologies aimed at improving traceability (Chen et al., 2021; Menon and Jain, 2024).

3.3 Logistics and infrastructure risks

Dependence on logistics and infrastructure can put companies at risk in terms of storage issues, transportation problems, and port troubles. Such problems can severely disrupt the food distribution network along with the performance of the company and its processes from supplier to customer (Ali and Govindan, 2021; Abu Nahleh et al., 2023). The outbreak of COVID-19 led to issues with transportation, inadequate storage facilities, food loss, and a shortage of labor, all of which impacted the availability, access, utilization, and stability of food. These disturbances not only disrupted food production and supply but also compromised food quality, freshness, hygiene, and cost-efficiency, exacerbating food insecurity in both the Global North as well as Global South (Abu Nahleh et al., 2023). Incorporating Industry 4.0 technology (I4T) can offset the negative impacts of operational risks affecting the performance of agricultural food distribution networks (Ali and Govindan, 2021). Digital transformation minimizes operational risk such as in COVID-19. In South Asia, the research conducted on logistics operation, more especially in transportation and warehousing functions, can enhance supply chain resilience (Umar and Wilson, 2023).

3.4 Labor and workforce risks

Seasonal and migrant worker shortages coupled with strikes have heavily affected all links of AFSC (Kuizinaïtè et al., 2023). Human workforce is very important for smooth functioning of supply chain. During periods of labor shortages, Canada's peak agricultural season and its reliance on temporary immigrant worker programs may result in disruptions, food losses, and reduced competitiveness (Preibisch, 2010). Alterations in immigration labor policies have afforded employers more sway in terms of staffing rights. This serves to underscore the value of immigrant labor in tackling shortages and sustaining productivity and stability in the labor market. As per Abu Nahleh et al. (2023), food losses and logistics disruptions occurred along the food value chain due to labor shortages, increased transportation costs, and limited storage capacity during the COVID-19 pandemic. The deficiencies in AFSC have unmasked their vulnerabilities and their dependence on a flexible labor strategy like digitization and automation (Ali and Govindan, 2021; Dadi et al., 2021).

3.5 Political and regulatory risks

An export ban or regulatory changes are the main risks that can disrupt AFSC. It can disrupt the global supply chain of food and agriculture. Most importantly, it can impact countries that lack self-sufficiency in agri-food (Kuizinaïtè et al., 2023). According to Koppenberg et al. (2020), more than 20 countries have AFSC disruptions. Unanticipated governmental rules and regulatory changes could threaten the agri-food distribution. Because of coronavirus

disrupting the border and food trade, better policy measures could limit the number of interruptions. The measures may relate to border delays, certification and food trade rules (Deconinck et al., 2020). The agri-food distribution faces major risk standards due to government policy reforms and compliance measures (Kuizinaïtè et al., 2023). Excessive and inadequate regulation can harm the agricultural food sector, as excessive regulation may limit innovation and competitiveness, while insufficient regulation can lead to instability, including food safety issues, fraud, and unethical behavior (Zhao et al., 2017; Kuizinaïtè et al., 2023). Changes in regulations can disrupt the flow of goods within the supply chain, leading to delays and inefficiencies. During the COVID-19 pandemic, sudden border and certification rule changes led to shipment delays and spoilage of perishable agri-food products. Adjusting a company's production, package, or even the bus one of those cargo ships or trucks is likely to be costly. As per the study by Srivastava and Dashora (2021) and Menon and Jain (2024), to alleviate those risks, the policymakers and the stakeholders within the industry should look for innovative ways to achieve a balance between effectiveness and security. This can be achieved with the help of electronic traceability technology and blockchain, through which transparency and compliance can be enhanced.

3.6 Technological and cyber risks

Researchers have proposed adopting digital monitoring tools and innovations (such as AI, IoT and blockchain) to improve the traceability, openness, and efficiency of agricultural food distribution networks, which cyber-attacks could significantly compromise (Dadi et al., 2021; Ali and Govindan, 2021). According to Dadi et al. (2021), automation and digital platforms may help in handling issues concerning food safety and traceability but might also cause new risks. Tech platforms such as IoT, blockchain, and AI can help simplify processes and combat vulnerabilities and setbacks within the agricultural food distribution network (Ali and Govindan, 2021). For example, digital traceability systems can boost efficiency, support food protection, and increase openness throughout the distribution network (Charlebois et al., 2024). Digital twins can help create more transparency, reduce lead time, respond to unforeseen events, and make better use of current practices and resources. However, these technologies also bring new risks. Adoption of Industry 4.0 requires skilled workers, funding and economic benefits. Agricultural food supply chain can also face data privacy and cybersecurity threats from digital platforms (Saha et al., 2024). Technological solutions in agriculture rely heavily on global supply chains, which can create new vulnerabilities. Many of the components for technologies like IoT, blockchain, and AI come from far-off, expensive suppliers in the Global North. This dependence makes these systems prone to disruptions, which, honestly, raises questions about their long-term reliability and security. Moreover, sustainability could also be a big issue.

3.7 Social and consumer risks

The agricultural food supply chain is significantly affected by changes in consumer demands—for example, the growing preference for organic, plant-based, and sustainable foods. These changes require

TABLE 1 Summary of economic vulnerabilities in agri-food systems.

Vulnerability	Description	Affected region(s)	References
Smallholder financing constraints	Limited access to credit, insurance, and technology	Sub-Saharan Africa, South Asia	Teye and Quarshie (2021) and Amarnath et al. (2023)
Lack of crop insurance	Vulnerability to climate and extreme weather without risk mitigation tools	Developing countries	Ceballos et al. (2019) and Amarnath et al. (2023)
Digital divide	Low adoption of digital agriculture due to infrastructure, cost, and skills	Sub-Saharan Africa, South Asia	Gumbi et al. (2023) and Teye and Quarshie (2021)
Supply chain disruptions (COVID-19)	Transport and demand shocks exposed farmer vulnerability	Sri Lanka, Global	Ortmann and King (2010) and Zhao et al. (2022)
Export dependency	Disruption in global demand/supply chains due to external shocks	Ghana (cocoa), Malaysia (palm oil)	Tan et al. (2023) and Grabs and Carodenuto (2021)
Rising food prices	Reduced household intake of calories and nutrients	India, Global South	Bouis et al. (2011) and Vellakkal et al. (2015)
Poor food access during crises	Supply chain breaks lead to food insecurity, especially for vulnerable groups	Low-income, female-headed households globally	Haddad (2000) and Welch and Graham (2000)
Fertilizer and grain trade disruptions	War-related supply cuts lead to rising input costs and food insecurity	Africa, Middle East, Global	Liu L. et al. (2023) , Hatab (2022) , and Hussein and Knol (2023)
Natural disaster shocks	Climate-induced events (droughts, floods) damaging crops and infrastructure	East Africa, Pakistan, Australia	Nhamo et al. (2019) , Verschuur et al. (2021) , and Tirivangasi (2018)
Informal/local system neglect	Underrepresentation of informal food systems in research and policy	Developing countries	Berti and Mulligan (2016) and Sid et al. (2021)

Note: This table reflects the growing vulnerability of food production systems arising from dependence on imported agricultural inputs, particularly fertilizers, which link domestic food security to global market dynamics.

adjustments in production, processing, and distribution systems, supported by technologies such as blockchain to improve tracking and transparency ([Bhat et al., 2021](#); [Toader et al., 2024](#)). Shifting consumer demands are driving changes in the AFSC. To remain competitive and respond to evolving market needs, organizations must adopt digital technologies and sustainable practices ([Ali and Govindan, 2021](#); [Krstic et al., 2023](#)). Food recalls caused by contamination or labeling errors negatively affect consumer trust and brand reputation. Surveys show that consumers are particularly concerned about recalls involving gluten-free products ([Liu S. et al., 2023](#)). However, when companies manage recalls in a transparent and efficient way, they can maintain consumer confidence. Corporate social media can help reduce the impact on stock prices, which tend to fall more sharply after a first recall than after multiple recalls ([Wang et al., 2002](#); [Lee et al., 2015](#)).

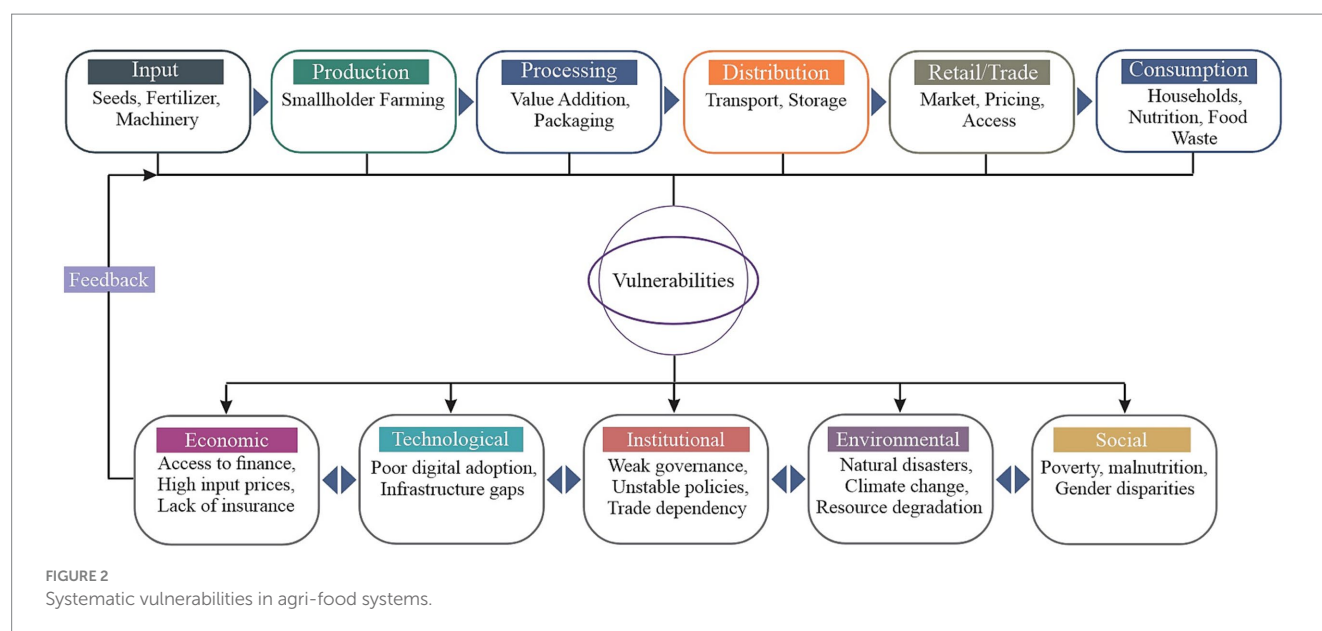
4 Economic vulnerabilities in AFSs

4.1 Vulnerabilities of smallholder farmers

Small farmers are economically vulnerable due to constraints on financing, insurance, and technology, as presented in [Table 1](#). In addition, dependence on imported agricultural inputs such as mineral fertilizers, agrochemicals, and machinery exposes both Global South and Global North countries to international price and supply fluctuations. These dependencies increase production costs and food insecurity risks during global crises ([Hatab, 2022](#); [Liu L. et al., 2023](#); [Hussein and Knol, 2023](#)). Stringent collateral demands, high interest rates, and limited access to affordable credit imposed by financial institutions severely hinder their ability to invest in new production methods, risk management strategies, and modern technologies ([Teye and Quarshie, 2021](#); [Amarnath et al.,](#)

[2023](#)). This lack of credit channels restricts farmers from investing in modern technologies needed to boost productivity and reduce food insecurity ([Teye and Quarshie, 2021](#)). The majority of small-scale farmers do not have access to low-cost crop loss insurance, leaving them heavily impacted by extreme weather events. [Gumbi et al. \(2023\)](#) stated that small farmers could benefit from digital agriculture to solve their problems. However, it is not widely adopted because digital systems are weak, digital skills are low, costs are high, and infrastructure is poor. Using image-based insurance (PBI) can improve coverage for small farmers. Policies, improved rural infrastructure, and combined solutions like weather insurance, agricultural advice, and climate-resistant seeds should be implemented to enhance smallholder farmers' capacity to cope ([Ceballos et al., 2019](#); [Teye and Quarshie, 2021](#); [Amarnath et al., 2023](#)). Smallholder farmers face rising costs due to dependence on imported inputs and soil degradation. This economic fragility are compounded by price volatility, unsustainable practices, and lack of financial support, highlighting the need for systemic policy changes.

Small farmers are very vulnerable in the agricultural food system due to their low resource base, high transaction costs, and operational compliance problems, as presented in [Figure 2](#). This figure presents an overview of AFS, showing sequential stages, input, production, processing, distribution, retail/trade, and consumption, along with their interconnected vulnerabilities. The middle section highlights cross-cutting risks, while the lower part classifies them into five dimensions, viz., economic, technological, institutional, environmental, and social. Economic and technological issues arise from limited finance and weak digital adoption, whereas environmental and social challenges like climate change, resource degradation, poverty, and gender disparities intensify vulnerabilities across stages. The coronavirus outbreak revealed this problem, especially in Sri Lanka, as a result of transportation and demand disruptions, which worsened



economic conditions (Ortmann and King, 2010). Small farmers without financial buffers were hard hit by external shocks. Farmers help build overall resilience capacity of AFSC, but they tend to be viewed as the least resilient ones in AFSC (Zhao et al., 2022). We need to tackle the susceptibility of small farmers in order to ensure their resilience and the sustainability of AFS. Strategies like developing alternative food networks, promoting collective action, and using digital technology can help reduce these vulnerabilities (Ortmann and King, 2010; Belhadi et al., 2024). As per Karunarathna et al. (2023), supply of agricultural inputs, loans, and subsidies by the government may lead to stabilization and sustaining of small farmer operations.

4.2 Export-oriented economies

AFSC depends on global markets for its inputs and outputs, which makes domestic economies more vulnerable. This can be seen in Ghana's cocoa and Malaysia's palm oil sectors. The pandemic has shown these weaknesses in the sector (Tan et al., 2023). Because of this reliance on global markets, food supply and demand might become unbalanced. At the same time, that reliance can create opportunities for resilience. Traders are critical to linking corporate sustainability targets to real-world impacts in food supply chains (Grabs and Carodenuto, 2021). Through better global connections, risks coming from market dependency can be reduced. To address the challenges of AFSC dependency, countries should develop supply chains with resilience by innovation, early warning mechanism and a middle path between global participation and domestic food security measures (Manikas et al., 2022; Joshi et al., 2023). The financial and industrial limits in developing regions make it hard to implement advanced infrastructure. These areas face challenges not just with limited capital and outdated infrastructure, but also with a shortage of skilled labor and the time-consuming need to train the workforce to manage new technologies properly. Furthermore, applying digital technologies including approaches such as 'Agri-Food 4.0' can contribute to lowering food waste, enable real-time tracking, and support scalability (Dadi et al., 2021). Using IoT, big data, and blockchain can assist in establishing traceability and enhancing the efficiency of AFSC systems in developing regions. However, a major challenge for

many countries is securing substantial investment for infrastructure development, especially in rural areas with limited access to energy, internet, and logistics, compounded by a lack of financial resources and technical expertise (Bhat et al., 2021; Ahmadzadeh et al., 2023).

4.3 Impact on food prices and household budgets

As highlighted in the introduction, food security encompasses not only the availability of food but also its access, stability, and utilization. The rise in food costs has impacted family budgets and nutritional stability, especially for vulnerable groups in low-income nations. A 50% hike in food price leads to a reduction of 5–15% in energy intake and a cut of 10–30% in iron intake, depending on income effects (Bouis et al., 2011). In India, the rise in food costs brings a higher likelihood of children becoming malnourished. The rate of emaciation in children, for example, rose from 18.8% in 2006 to 28.0% in 2009 (Vellakkal et al., 2015). When shortages arise, families switch to cheaper brands, purchase larger quantities and plant vegetable gardens. Shortages can affect food quality and variety, especially for the poor and female headed households. As noted by Welch and Graham (2000), malnutrition is not only a result of agricultural inefficiencies but also of poverty, which is a critical barrier to achieving food security. It is necessary to act on these angles simultaneously to give us food security and nutrition (which includes health) through agricultural production and food systems (Welch and Graham, 2000; Haddad, 2000).

5 Mitigation and risk management approaches

5.1 Policy and governance solutions

The agricultural food supply chain can experience reduced risk and disruption through policy and governance solutions, provided by the farm, Table 2. According to Charlebois et al. (2024), food security, regulatory frameworks, emergency plans, and digital

traceability are possible through national food strategies and trade agreements. OECD countries have adopted digital traceability systems to enhance supply chain performance, strengthen food safety, and boost transparency, while simultaneously reinforcing the robustness of disadvantaged SMEs (Ali et al., 2022; Charlebois et al., 2024). The food distribution network functions effectively amid the Covid-19 pandemic, meaning cross-border trade agreements are essential to ensure continuous food flow during the crisis (Hobbs, 2021). Geopolitical occurrences could interrupt agricultural supply chains, highlighting the necessity for robust trade agreements (Belhadi et al., 2024). As per Hong et al. (2023), the government adversely impacts global supply chain performance through trade-related policies and non-tariff measures. Similarly, they promote stability and resilience through research and development incentives, domestic demand stimulation, and subsidy design. Local food systems, including smallholder farms and community networks, help with resilience by being more flexible and not relying so much on global supply chains during disruptions. Technology like blockchain help adaption, while supportive policies strengthen resilience at both the levels (Quayson et al., 2020; Trivellas et al., 2020; Rejeb et al., 2022).

5.2 Financial instruments and risk transfer

Financial instruments and risk-sharing methods such as agricultural insurance, micro-credit, and weather-indexed insurance can help reduce risks linked to AFSC. According to Al-Maruf et al. (2021), climate-linked crop insurance may lessen the effects of climate-related risks, as demonstrated in a pilot project in Bangladesh. This approach connects the cost of disasters to weather variables and is a better risk transfer mechanism than standard crop insurance. On the contrary, products based on indices improve the likelihood of accessing micro-credit by small agricultural producers and transferring the risk of disaster to the global market (Skees and Barnett, 2006). The effectiveness of different financial instruments, including the WIBCI, depends on the context and the state of their use. A limited amount of data, high cost and illiteracy of farmer insurance, etc., are the challenges (Al-Maruf et al., 2021). As per Ali and Govindan (2021), the use of Industry 4.0 technology in AFSC

can mitigate both operational and financial risks during conditions like the COVID-19. AFSC risk management strategy is an integrated set of financial tools, risk transfer mechanisms and technological means. How effective it is, depends on education, participation in additional strategies, and integration with a range of other resilience factors like traceability, collaboration, and agility (Mishra and El-Osta, 2002; Zhao et al., 2017). Future research should make cross-border comparisons and adopt a multi-sectoral approach to better understand risk and sustainable management in AFSCs (Zhao et al., 2024).

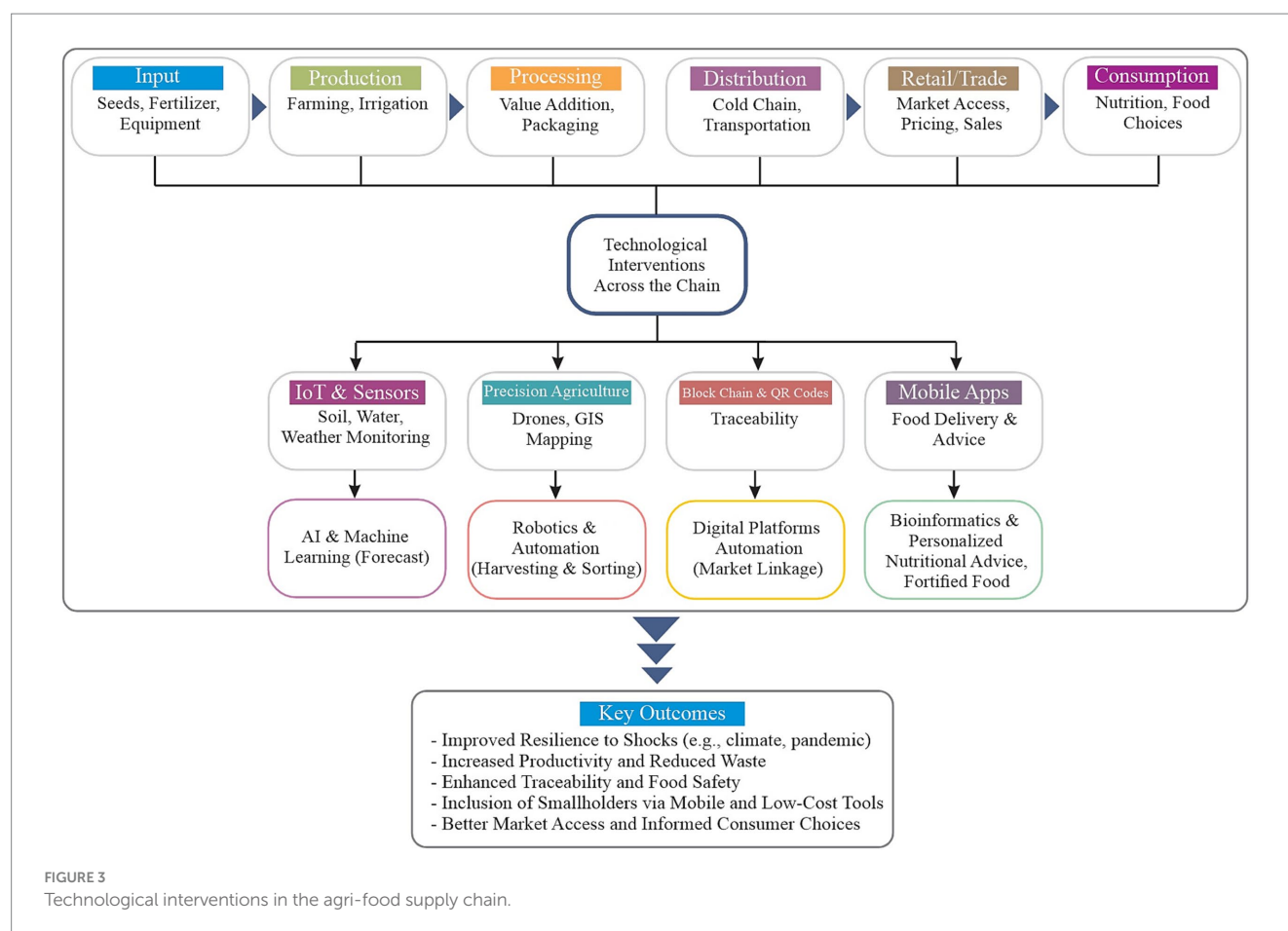
5.3 Technological innovations

AFSC is one of the strategic areas where blockchain and IoT as well as AI should be used to reduce risks, improve traceability and operation efficiency, as illustrated in Figure 3. This figure illustrates technological interventions across AFSC from input to consumption, showing the flow of goods and information. The middle section highlights key technologies such as IoT and sensors, precision agriculture using drones and GIS, blockchain and QR codes, and mobile apps supported by AI, robotics, and digital platforms that enhance forecasting, automation, and market connectivity. These technologies facilitate transparency, monitoring, forecasting, and inventory management in real-time (Rui and Sundram, 2024). AgriBlockIoT, a platform built on blockchain and IoT, aims at improving tracking in AFSC (Odimarha et al., 2024). Quick detection of contaminated items for precise recall helps improve food safety by reducing foodborne illnesses (Oriekhoe et al., 2024). But, the SMEs bear the burden of things like high costs, shortage of skilled labour, data security issues, and the employees' limited technical knowledge (Rui and Sundram, 2024; Odimarha et al., 2024). It is advised to run pilot projects, train workers, secure data, and get help from government and industry to help address these issues. The bottom section of Figure 3 summarizes the key outcomes of these technological applications, including improved resilience to shocks (e.g., climate or pandemic), increased productivity, reduced waste, enhanced traceability, and inclusion of smallholders through mobile and low-cost digital tools. By tackling these challenges, organizations can create a sustainable and effective AFSC, improving global market

TABLE 2 Comparison of risk mitigation tools in agri-food systems.

Tool	Type	Benefits	Limitations	Target users
Crop insurance	Financial/risk transfer	Provides compensation for yield loss due to natural disasters	High premium costs, low awareness, limited reach in developing regions	Smallholder farmers, agribusinesses
Weather-indexed insurance	Financial/risk transfer	Ties payouts to weather conditions, faster and less biased claims	Basis risk (payout may not match actual loss), data access and literacy issues	Small-scale farmers, rural cooperatives
Microcredit	Financial	Access to small loans for investment in inputs and technologies	High interest rates, collateral requirements, risk of over-indebtedness	Marginalized farmers, women entrepreneurs
Blockchain	Technological (traceability)	Improves transparency, traceability, and trust in the food supply chain	High cost, limited digital literacy, data security concerns	Exporters, logistics firms, regulatory bodies
Internet of things	Technological (monitoring)	Enables real-time monitoring of crops, logistics, and storage conditions	Expensive devices, poor internet access in rural areas, skills gap	Large farms, tech-oriented cooperatives

Sources: Ali and Govindan (2021), Al-Maruf et al. (2021), Skees and Barnett (2006), Rui and Sundram (2024), and Odimarha et al. (2024).



presence while also promoting long-term environmental, social, and economic sustainability (Rui and Sundram, 2024; Odimarha et al., 2024).

5.4 Local food systems and supply chain flexibility

By shortening supply chains, forming farmer cooperatives, and engaging in urban agriculture, local and regional food systems are strengthened with a view to increasing resilience and sustainability, lessening dependence on global networks, as well as enhancing nutritional stability in areas like the UAE (Sharma et al., 2023). The COVID-19 pandemic underscores that managing interruptions strategically starts with a careful assessment of options available in supply chain design and flexibility. Supplier diversification and multiple procurement are strategies that can help manage disruption effects by reducing the chances of reliance on a single source (Tomlin, 2006; Saghafian and Van Oyen, 2016). Effective inventory management strategies, like positional strategies, help reduce risks in the supply chain. Techniques such as Risk Exposure Index (REI) and Conditional Value at Risk (CVaR) can identify the optimal inventory layout (Gao et al., 2019). The transition from food services to food retail demands readjustments within the agricultural product supply network, since interruptions could occur (Tomlin, 2006; Hobbs, 2021). By investing in strategies like flexible sourcing and multiple delivery locations, supply chain issues can be resolved.

Reliability of AFSC can be further improved with supplier diversification. Also, strategic inventory management will help enhance agricultural food supply chain resilience. Building on the earlier points about the importance of flexibility and strategic management in supply chains, investing in automation, digitization, and online deliveries will be key to strengthening resilience and lessen dependence on global networks (Hobbs, 2021). These technologies can directly address the vulnerabilities as discussed before, like disruptions in supply chains, by boosting efficiency, enabling real-time monitoring, and increasing adaptability within agricultural food systems. These shifts, if properly implemented, might be a game-changer in tackling future disruptions.

5.5 Collaborative education and capacity building

The importance of public-private partnerships is to use resources, share infrastructure, and plan for the long-term disruptions a man-made or natural disaster can cause, to increase robustness as well as efficiency of AFSC (Gabler et al., 2017). It is especially important that there be public-private cooperation in urban water distribution design. Regional cooperation between public utilities can contribute to enhancing the reliability along with financial stability of the water provision (Gold et al., 2022). Collaborating with public authorities and private enterprises, alongside global cooperation and capacity building, may enhance food supply resilience by improving access to

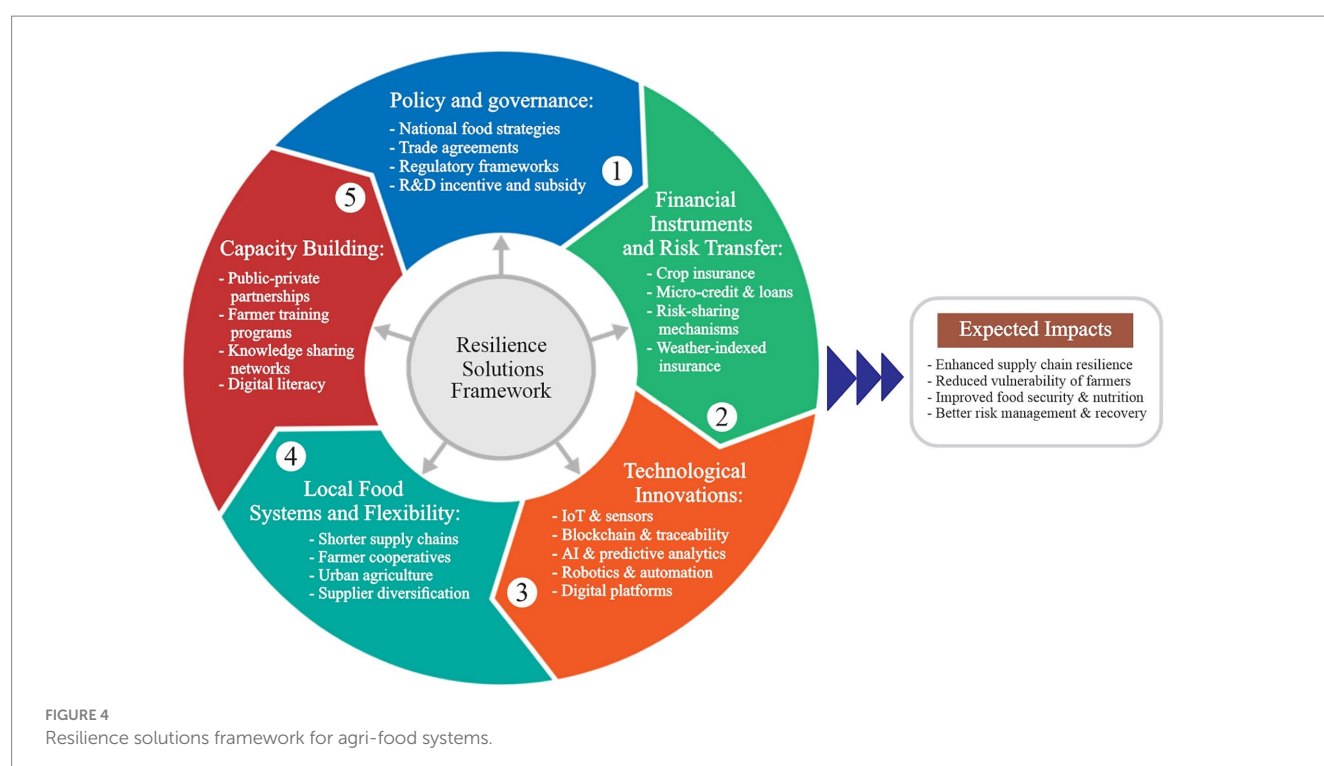
resources, knowledge, and innovations while reducing risks associated with AFSC. According to Zhao et al. (2022), developing training programs and coordinating the efforts of farmers, associations, and other stakeholders can improve adaptability and supply chain resilience. When farmers and workers interact regularly and share knowledge, they better handle challenges that could cause disruption. Digital technology also aids in the design of resilience strategies and uncertainty management in AFSCs (Belhadi et al., 2024). Figure 4 presents the resilience solutions framework comprising five pillars that enhance AFS resilience. These include policy and governance (strategies and regulations), financial instruments (insurance and credit), technological innovations (IoT, AI, blockchain), local food systems (shorter supply chains and diversification), and capacity building (training and partnerships). Together, they improve resilience, reduce vulnerability, and strengthen food security and risk management.

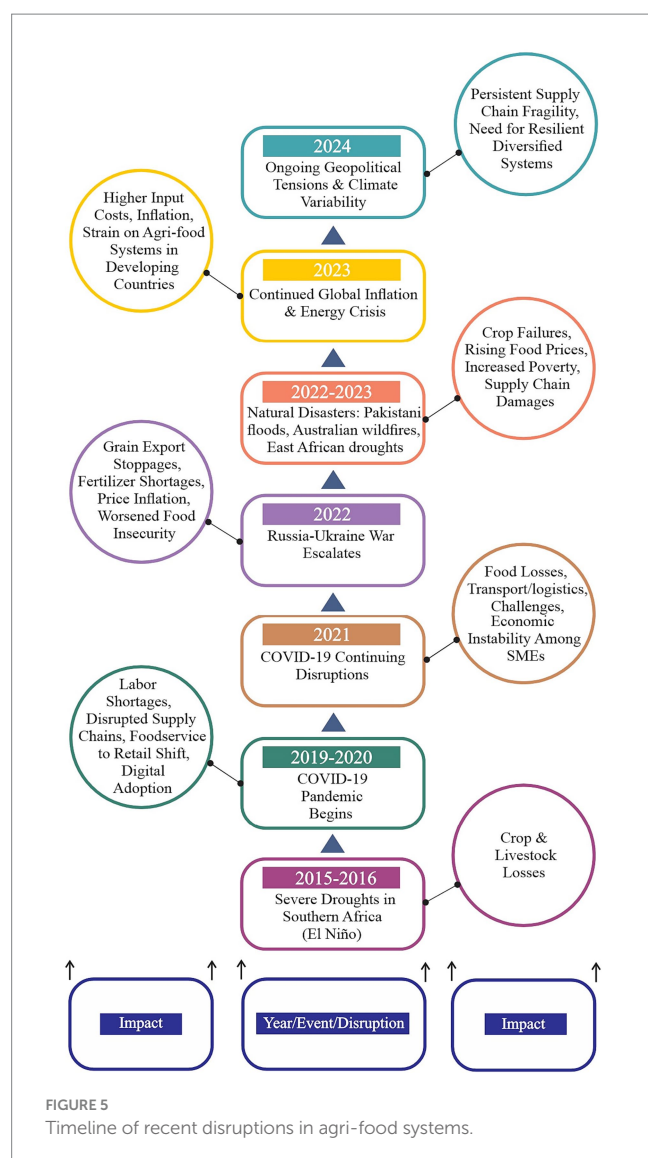
6 Case studies of recent disruptions

The COVID-19 pandemic caused labor shortages within meatpacking and produce cultivation due to employee infections of COVID-19 at manufacturing facilities and farms (Hobbs, 2021), as shown in Figure 5. This figure illustrates a timeline of key global disruptions from 2015 to 2024 and their cascading impacts on AFSC. Events such as El Niño droughts, COVID-19, the Russia–Ukraine war, and ongoing inflation are linked with impacts including food insecurity, rising prices, and supply chain fragility. The figure emphasizes how overlapping crises have compounded vulnerabilities, highlighting the urgent need for more resilient, adaptive, and diversified AFSs. Food losses occurred because of labor disruptions, higher transportation costs, and poor storage facilities during the

pandemic (Abu Nahleh et al., 2023). As dietary habits shifted from hospitality sectors to grocery retail, the associated supply chains faced supply, logistics, and distribution challenges that destabilized the economy, especially among SMEs (Hobbs, 2021; Ramanathan et al., 2021). The coronavirus pandemic caused adverse impacts on nutritional stability and income sources worldwide, particularly in areas that had weak supply chains, high poverty, and market informality before the pandemic (Swinen and Vos, 2021). The agricultural food sector fast-tracked the implementation of intelligent technologies and Industry 4.0 solutions during the worldwide health crisis, resulting in improved business processes and reduced operational risks (Ali and Govindan, 2021). Investment in online delivery infrastructure has permanently altered the food retailing landscape. The COVID-19 crisis exposed both the vulnerabilities as well as resilience of AFSC. It has led to not only a growth in online retailing of food and agricultural goods but also a rise in the use of digital technologies. Many believe these changes may last long, leading to greater automation and digitalization, and possibly more flexible supply chain configurations (Hobbs, 2021; Ali and Govindan, 2021; Suali et al., 2024).

Due to the war, pre-existing problems in AFSC have worsened, as illustrated in Table 3. Panic, inflation, and food insecurity have crippled import-dependent regions (Hussein and Knol, 2023). The halt in cereal shipments out of Russia and Ukraine has caused serious disruption to global grain trade, particularly wheat. Many countries have faced wheat supply shocks, with nearly 30 countries suffering. Fertilizer supply has also seen disruption, given that Russia is a major supplier of natural gas and fertilizers (Ben Hassen and El Bilali, 2022; Esfandabadi et al., 2022; Liu L. et al., 2023). According to Hatab (2022), the war affects the food supply chain via fuel sectors, transport corridors, farm supplies, rising food costs, and commercial restrictions. This has especially adverse effects on access to adequate





food and dietary health in Africa. As a result, many people are now struggling with poverty and hunger. Agricultural food networks and low-income communities are more exposed to increasing energy costs, and high food prices worsen malnutrition and dietary standards (Arndt et al., 2023). The war between Russia and Ukraine has significantly impacted worldwide food availability. Across Africa and other regions reliant on imports, the conflict underscores the vulnerability of food systems. We require more positive approaches toward improving the global food supply. Furthermore, creating resilient and diversified supply chains is necessary (Ben Hassen and El Bilali, 2022; Hatab, 2022).

Natural disasters like floods in Pakistan, droughts in East Africa, and wildfires in Australia damage agricultural food supply chains and result in significant economic and social losses. In 2015/2016, southern Africa experienced severe droughts associated with El Niño that destroyed crops and livestock, leaving around 40 million individuals requiring global support (Nhamo et al., 2019). Climate change-related disasters like wildfires and droughts can devastate agriculture, leading to crop failures, escalating food costs, heightened poverty, and inadequate access to nutrition, especially in trade-dependent countries with vulnerable agricultural systems (Verschuur et al., 2021). Natural calamities harm the economy in many ways. For example, it will affect agricultural production, market access, trade, and food supply. Also, it will negatively impact farm income and employment generation. Increased food insecurity and malnutrition is likely to impact the poorest communities most (Tirivangasi, 2018). After climate-related shocks damage agriculture, it is essential to have strong and resilient AFSC to ensure sustainable food systems. Measures to mitigate these risks could be to adopt the Industry 4.0 technology, make traceability systems, cooperate and agility, and invest in irrigated agriculture and more variety of crops (Zhao et al., 2017; Nhamo et al., 2019; Ali and Govindan, 2021). Adaptation interventions and policies that assist agricultural communities to establish resilient systems are crucially important (Tirivangasi, 2018; Nhamo et al., 2019).

TABLE 3 Classification of agri-food disruptions by event type.

Disruption type	Nature of impact	Region	Outcome	References
COVID-19 Pandemic	Labor shortages, supply chain disruptions, shift from food service to retail	Global; Sri Lanka; Developing nations	Food losses, higher transport costs, SME collapse, digitalization acceleration	Hobbs (2021), Abu Nahleh et al. (2023), and Swinnen and Vos (2021)
Russia-Ukraine Conflict	Grain and fertilizer supply shocks, inflation, trade route disruption	Africa, Middle East, Global Importers	Wheat shortages, rising fuel and food prices, increased poverty and hunger	Hatab (2022), Hussein and Knol (2023), and Liu L. et al. (2023)
Pakistan Floods	Massive crop and infrastructure destruction due to heavy rainfall	Pakistan	Agricultural and economic loss, food insecurity, displacement	Referenced indirectly in broader natural disasters context
East African Drought	Long-term drought reducing agricultural productivity and increasing food insecurity	East Africa	Food shortages, livestock deaths, international aid requirement for 40 million people	Nhamo et al. (2019) and Tirivangasi (2018)
Australian Wildfires	Fires damaging farms and supply chains	Australia	Crop destruction, loss of biodiversity, supply chain interruption	Verschuur et al. (2021) and Tirivangasi (2018)
Southern Africa El Niño Drought (2015/16)	El Niño-linked drought caused crop and livestock loss	Southern Africa	Millions requiring food aid, income and employment loss	Nhamo et al. (2019)

TABLE 4 Identified gaps in literature and future research directions.

Identified gap	Description	Future research direction	References
Limited focus on smallholder and informal food systems	Most research emphasizes industrial, formal, and global supply chains	Investigate local, informal, and regional food systems and their roles in resilience	Berti and Mulligan (2016) and Sid et al. (2021)
Underexplored digital innovation in local networks	Lack of research on digital tools in small/local agri-food systems	Study how blockchain, mobile apps, and IoT can improve traceability and trust in informal systems	Bhat et al. (2021) and Chiaraluce et al. (2024)
Weak interdisciplinary approaches	Supply chains studied in silos, ignoring the interconnected nature of agri-food systems	Employ systems thinking and interdisciplinary methods (economics, agriculture, tech, sociology, environment)	Horton et al. (2017)
Inadequate analysis of local response to global disruptions	Local systems' resilience during crises like COVID-19 not well-documented	Compare responses of informal/local vs. formal/global supply chains under disruption scenarios	Sid et al. (2021) and Hobbs (2021)
Lack of comparative regional studies	Most studies focus on single-country or single-region cases	Conduct cross-border and comparative research on resilience, risk management, and innovation adoption	Zhao et al. (2024) and Ali and Govindan (2021)
Minimal assessment of technology adoption barriers for SMEs and smallholders	Challenges like cost, digital illiteracy, and infrastructure gaps remain understudied	Explore inclusive strategies for capacity building and scalable digital infrastructure for marginalized groups	Rui and Sundram (2024) and Odimarha et al. (2024)

7 Discussion

Despite intensity along with extent of studies on the agro-food distribution network, several important challenges remain underexplored, as shown in Table 4. Most past studies looked at big supply chains, like ones involving trade, big farms, and advanced tech. Most of these studies look at themes like sustainability, digital innovation and adaptability within industrial scale distribution systems (Dadi et al., 2021; Rejeb et al., 2021; Ali et al., 2022). Local and informal food networks, though often overlooked, are very important for food access and livelihoods, especially in both developing and industrialized countries. In the Global North, these systems face big challenges like land degradation and fewer farmers, making them more reliant on big supply chains. By strengthening local food systems, we can make them more resilient to global disruptions. Globalization has made everyone vulnerable, showing that local supply chains within a global framework are essential for food security. Berti and Mulligan (2016) observe how regional food centers assist connecting small-scale farmers to customers and build better food supply chains. These local systems help promote the concept of shared value, as well as contribute to the existence of small farms in a very competitive environment (Jia et al., 2024). This topic has not been studied enough. Therefore, further investigation will help us learn how these local food systems work and how to make them stronger.

Future studies should compare informal and traditional food systems with formal suppliers chain. This refers to informal systems and facilities that fill the gaps. Researchers can analyze how a local food system responds to crises like the COVID-19 outbreak, that, while disrupting the broader global distribution network, demonstrated the local market's potential resilience (Sid et al., 2021; Chavez-Miguel et al., 2024). Another important area of research refers to employing digital tools (e.g., blockchain, mobile applications) to enhance trust, tracking, and openness within informal food networks (Bhat et al., 2021; Chiaraluce et al., 2024). Researchers

should further explore the broader impacts of local food systems on environment, economy, and society, especially in rural areas. Interdisciplinary methods are urgently needed to study the agricultural food supply chain thus improving it. The systems are complex in nature and comprise many interrelated components, from land use and food production to food processing, transport and consumption. Systems must always be treated holistically by researchers, as there are many interactions and feedback loops between system components (Horton et al., 2017; Basset, 2024). As per Dadi et al. (2021), the above technologies can help in lessening food waste, improving efficiency as well as expanding production scale.

8 Conclusion

AFS faces many economic vulnerabilities that threaten nourishment availability and the reliability of distribution networks. Smallholder farmers rank among those hardest hit because of restricted availability of finance, insurance, and modern technology (SDG 1 and SDG 2). Export-dependent economies also face risks when global markets are disrupted, as seen during the coronavirus outbreak and the Russia-Ukraine war. Increasing food costs persist in impacting economically disadvantaged families, limiting their capacity to purchase healthy food. To address these challenges, several solutions can be applied. These include better policies, investment in digital technologies (SDG 9), improved infrastructure, and access to financial tools such as crop insurance and microcredit. Supporting local food systems, building flexible and shorter supply chains, and encouraging cooperation between governments, private sectors, and farmers are also important. Recent crises have shown the necessity of developing robust AFSs capable of adapting to disruptions and recovering quickly. Future strategies must focus on education, innovation, and collaboration to strengthen the

entire food supply chain. In this way, we are able to guarantee a more secure, sustainable (SDG 12), as well as fair food system for all, while enhancing resilience to climate-related risks (SDG 13).

Author contributions

YX: Writing – original draft, Investigation, Data curation. JY: Supervision, Writing – original draft, Funding acquisition. MM: Conceptualization, Writing – review & editing, Data curation. AM: Investigation, Resources, Writing – review & editing.

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